Dynamical Systems of Particles Around a Kerr Black Hole

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Introduction

- This report investigates the dynamics of particles orbiting a Kerr black hole.
- By employing Hamiltonian and Hamilton–Jacobi formalisms, we derive key conserved quantities and analyze the motion of test particles in curved spacetime.
- Numerical simulations further illustrate complex phenomena, offering insights into high-energy astrophysical processes.

Basic Settings

- Kerr Black hole has stationary and axisymmetric properties.
- We adopt the Boyer-Lindquist coordinates (t, r, θ, ϕ)
- M is the black hole mass in geometrized units ($c = G = 1$).
- spin (or rotation) parameter is defined as $a = \frac{J}{M}$ where J is the total angular momentum.

Metric of Kerr Black Hole

$$ds^2 = -\frac{\Delta - a^2 \sin^2\theta}{\Sigma} dt^2 - 2a \sin^2\theta \frac{r^2 + a^2 - \Delta}{\Sigma} dt\, d\phi + \frac{\Sigma}{\Delta} dr^2 + \Sigma\, d\theta^2 + \frac{(r^2 + a^2)^2 - \Delta\, a^2 \sin^2\theta}{\Sigma} \sin^2\theta\, d\phi^2 \quad (1)$$

where Σ and Δ are defined as

$$\Sigma = r^2 + a^2 \cos^2\theta \quad (2)$$

$$\Delta = r^2 + a^2 - 2Mr \quad (3)$$

The Ergoregion and Horizons

- The Kerr black hole has an ergosphere where frame dragging prevents any object from remaining stationary.

$$r_{\text{ergosphere}} = M + \sqrt{M^2 - a^2 \cos^2 \theta}$$

- There are two horizons:

$$r_+ = M + \sqrt{M^2 - a^2} \quad \text{(outer event horizon)}$$

$$r_- = M - \sqrt{M^2 - a^2} \quad \text{(inner Cauchy horizon)}$$

- Additionally, there is a ring-shaped singularity.

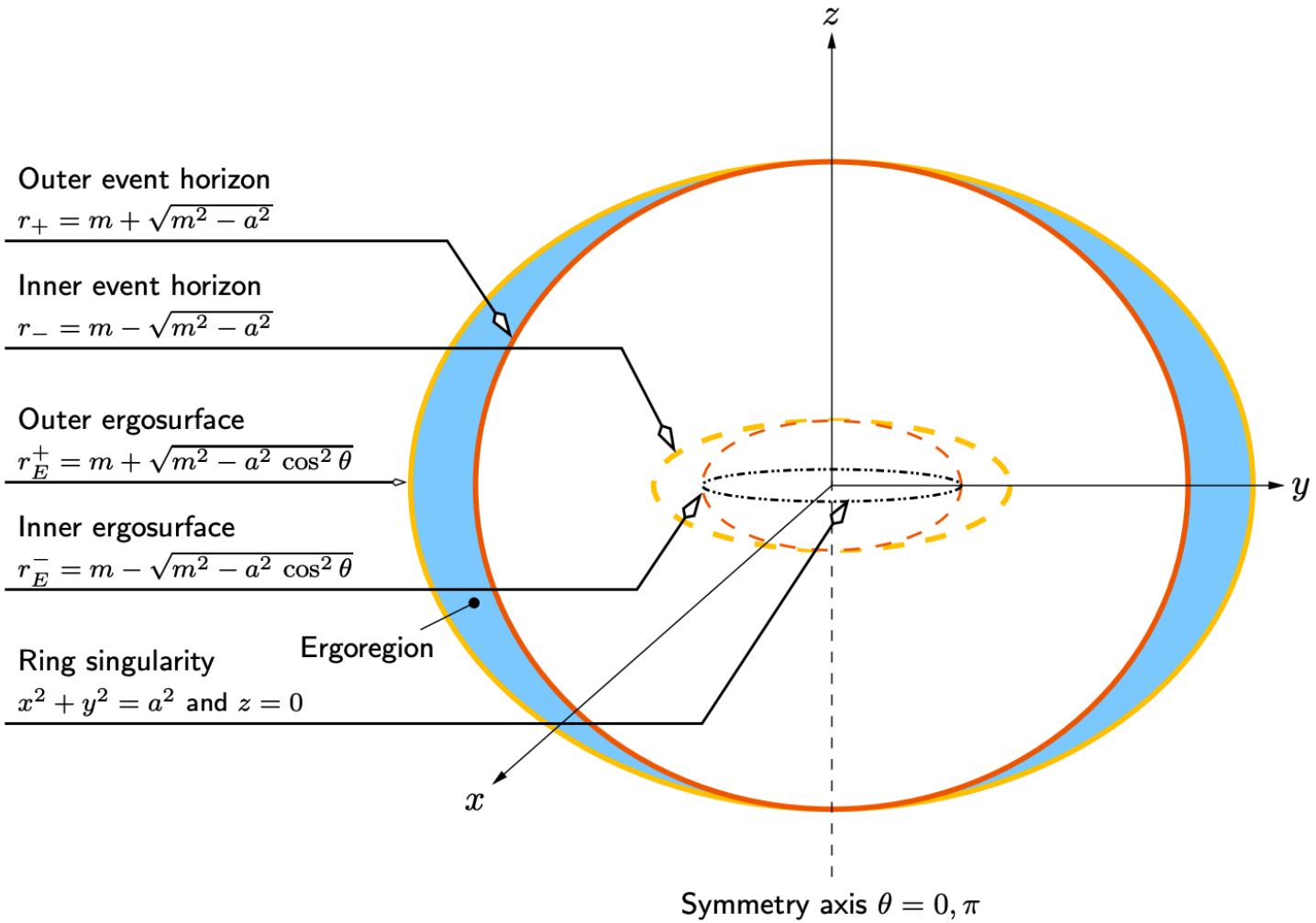


Figure: Schematic Kerr Black Hole [4]

Three Constants

As always, for a geodesic with affine parameter λ, we use x to denote the position of the geodesic, just keep in mind that we are working with Boyer-Lindquist coordinates. We use u to denote the velocity vector.

Definition

The *Lagrangian* is defined as:

$$\mathcal{L}(x, \dot{x}) := \frac{1}{2} g(\dot{x}, \dot{x}) = \frac{1}{2} g_{\mu\nu} \dot{x}^\mu \, \dot{x}^\nu .$$

The *conjugate momentum* is given by:

$$p_\mu := \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} = g_{\mu\nu} \, \dot{x}^\nu .$$

Notice that the physical momentum $p^\mu = u^\mu$, so lowering the index we have $p_\mu = g_{\mu\nu} u^\nu$ which is just the conjugate momentum defined above.

Lemma

p_t and p_ϕ are constant along the geodesics.

Pf: Stationary $\Longrightarrow$ Killing vector $k = (1, 0, 0, 0)$.
Axisymmetric $\Longrightarrow$ Killing vector $m = (0, 0, 0, 1)$ (wlog, assume it rotates around the z-axis)
So consider how $\xi_\mu p^\mu$ changes along the geodesic, we can show

$$p^\nu \nabla_\nu (\xi_\mu p^\mu) = 0 \tag{4}$$

- The energy and angular momentum $E = -k^\mu u_\mu = -p_t$ and $L = m^\mu u_\mu = p_\phi$ are constant along the geodesic.
- We should not forget there is another trivial invariant κ which is the invariant mass of the test particle, defined by

$$\kappa = g_{\mu\nu} u^\mu u^\nu = g_{\mu\nu} p^\mu p^\nu \tag{5}$$

$$\kappa = \begin{cases} -1 & \text{for timelike geodesics,} \\ +1 & \text{for spacelike geodesics,} \\ 0 & \text{for null geodesics.} \end{cases}$$

Hamilton–Jacobi Formalism

Definition

The Hamiltonian is defined as:

$$H(x,p) := p_\mu \dot{x}^\mu - \mathcal{L}(x,\dot{x}) = \frac{1}{2} p_\mu g^{\mu\nu} p_\nu.$$

Theorem (Jacobi)

Let $S(x,\lambda)$ solve the Hamilton–Jacobi equation

$$H\Big(x, \frac{\partial S}{\partial x}\Big) + \frac{\partial S}{\partial \lambda} = 0 \tag{6}$$

If $x(\lambda)$ satisfies $\dot{x}^\mu = \frac{\partial H}{\partial p_\mu}\big(x, \frac{\partial S}{\partial x}\big)$, then $x(\lambda)$ is a geodesic, and along it

$$\frac{\partial S}{\partial x^\mu} = p_\mu \quad \frac{\partial S}{\partial \lambda} = -H \tag{7}$$

- Since H does not depend explicitly on λ, t, ϕ, we look for solutions of the form (an ansatz)

$$S = -\tfrac{1}{2}\,\kappa\,\lambda \;-\; E\,t \;+\; L\,\phi \;+\; S^{(r\theta)}(r,\theta) \tag{8}$$

  where $S^{(r\theta)}$ is a function of r and θ to be determined.
- We follow the idea of B. Carter [5], we want to find a separable solution, so we further assume S has the following expression

$$S \;=\; -\tfrac{1}{2}\,\kappa\,\lambda \;-\; E\,t \;+\; L\,\phi \;+\; S^{(r)}(r) \;+\; S^{(\theta)}(\theta) \tag{9}$$

Definition

$$Q := p_\theta^2 - \cos^2\theta\left[(\kappa + E^2)a^2 - \frac{L^2}{\sin^2\theta}\right].$$

The Fourth Constant: Carter Constant

Theorem

Q defined above is also constant along geodesics, it is called the Carter constant.

Pf: Substitute the above assumption of S into the H-J equation, multiply by Σ, rearrange the equation such that the LHS only involves r and the RHS only involves θ:

$$\text{LHS} = \Delta\left(\frac{dS^{(r)}}{dr}\right)^2 - \kappa r^2 - \frac{(r^2+a^2)^2}{\Delta}E^2 + \frac{4Mra}{\Delta}EL - \frac{a^2L^2}{\Delta} \tag{10}$$

$$\text{RHS} = -\left(\frac{dS^{(\theta)}}{d\theta}\right)^2 + \kappa\, a^2\cos^2\theta - a^2\sin^2\theta E^2 - \frac{1}{\sin^2\theta}L^2 \tag{11}$$

In order to make the RHS$= -Q$, we add $a^2E^2 + L^2$ to both sides, so the equation becomes

$$\Delta\left(\frac{dS^{(r)}}{dr}\right)^2 - \kappa r^2 - \frac{(r^2+a^2)^2}{\Delta}E^2 + \frac{4Mra}{\Delta}EL - \frac{a^2L^2}{\Delta} + a^2E^2 + L^2 = -Q \tag{12}$$

Geodesic Equations

Lemma

We define the functions $R(r)$ and $\Theta(\theta)$ as:

$$R(r) = \Delta\Big[\kappa r^2 - (L - aE)^2 - Q\Big] + \Big[E(r^2 + a^2) - La\Big]^2 \tag{13}$$

$$\Theta(\theta) = Q + \cos^2\theta\Big[(\kappa + E^2)a^2 - \frac{L^2}{\sin^2\theta}\Big] \tag{14}$$

Then we have the following two ODEs:

$$p_\theta^2 = \left(\Sigma\dot{\theta}\right)^2 = \Theta(\theta), \quad p_r^2 = \left(\frac{\Sigma}{\Delta}\dot{r}\right)^2 = \frac{R(r)}{\Delta^2} \tag{15}$$

Pf: $p_\theta^2 = \Theta(\theta)$ follows immediately after the expression of Q. p_r is by the other expression of the Carter constant.

Theorem

The geodesics of a test particle around the Kerr black hole can be expressed as:

$$\dot{\theta} = \pm \frac{1}{\Sigma} \sqrt{\Theta}, \qquad \dot{r} = \pm \frac{1}{\Sigma} \sqrt{R} \tag{16}$$

$$\dot{\phi} = -\frac{1}{\Sigma}(aE - \frac{L}{\sin^2 \theta}) + \frac{a}{\Delta\Sigma}[E(r^2 + a^2) - aL] \tag{17}$$

$$\dot{t} = -\frac{1}{\Sigma}(a^2 E \sin^2 \theta - aL) + \frac{r^2 + a^2}{\Delta\Sigma}[E(r^2 + a^2) - aL] \tag{18}$$

Pf: $\dot{\theta}$ and $\dot{r}$ follow from the above lemma. The other two can be shown by solving

$$\begin{cases} p_t = g_{tt}\dot{t} + g_{t\phi}\dot{\phi} = -E \\ p_\phi = g_{\phi t}\dot{t} + g_{\phi\phi}\dot{\phi} = L \end{cases} \tag{19}$$

Numerical Method

- Mino time $d\lambda = \Sigma\, d\sigma$ [7] decouples the motion:

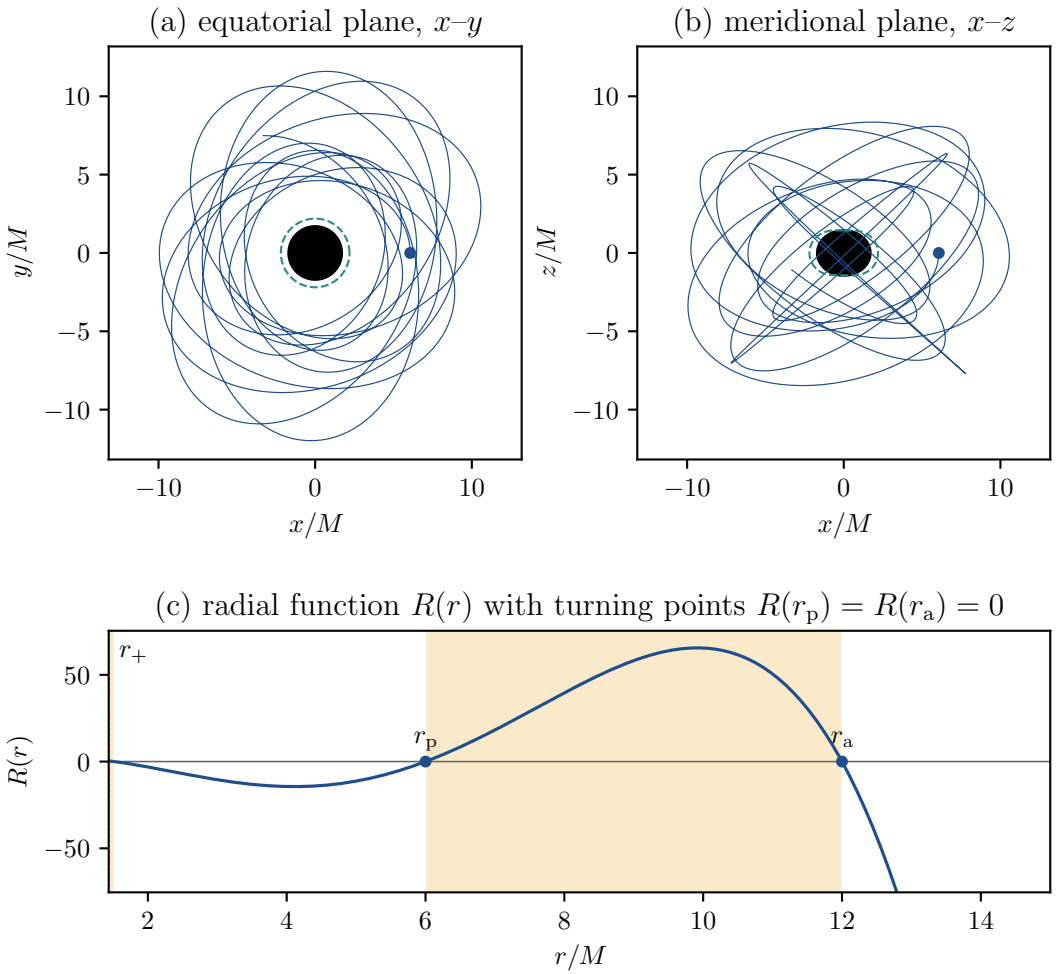
$$\left(\frac{dr}{d\sigma}\right)^2 = R(r), \qquad \left(\frac{d\theta}{d\sigma}\right)^2 = \Theta(\theta)$$

- We integrate the derivatives, smooth at turning points:

$$\frac{d^2 r}{d\sigma^2} = \tfrac{1}{2} R'(r), \qquad \frac{d^2 \theta}{d\sigma^2} = \tfrac{1}{2} \Theta'(\theta)$$

  together with $d\phi/d\sigma$, $dt/d\sigma$ from the theorem above.
- An orbit is chosen by its turning points $r_{\mathrm{p}}, r_{\mathrm{a}}$ and inclination ι: Newton's method solves $R(r_{\mathrm{p}}) = R(r_{\mathrm{a}}) = 0$ for E, L, Q.
- Adaptive Dormand–Prince Runge–Kutta [8]; first integrals hold to 10^{-9}; matches Hamilton's equations to 3×10^{-9}.

Example: $a = 0.9$, $r_{\mathrm{p}} = 6$, $r_{\mathrm{a}} = 12$, $\iota = 45^\circ$



Periapsis advance plus Lense–Thirring precession of the orbital plane. Interactive version: `kerr_geodesics.html`.

Extension to Charged Particle Dynamics

- Charged particle in a stationary, axisymmetric dipole magnetic field $A_\mu(r,\theta)$ on a fixed Kerr background [9]:

$$H = \tfrac{1}{2} g^{\mu\nu}(\pi_\mu - qA_\mu)(\pi_\nu - qA_\nu)$$

- t and ϕ are still cyclic, so E, L and κ are still conserved.
- But A_μ couples r and θ: the Hamilton–Jacobi equation no longer separates, there is no Carter-type fourth constant, and the poloidal motion can be chaotic.
- Orbits combine gyration around field lines, bouncing along them, and toroidal drift; the next slide shows two trajectories for $a = 0.9M$ that differ only in the injection angle.

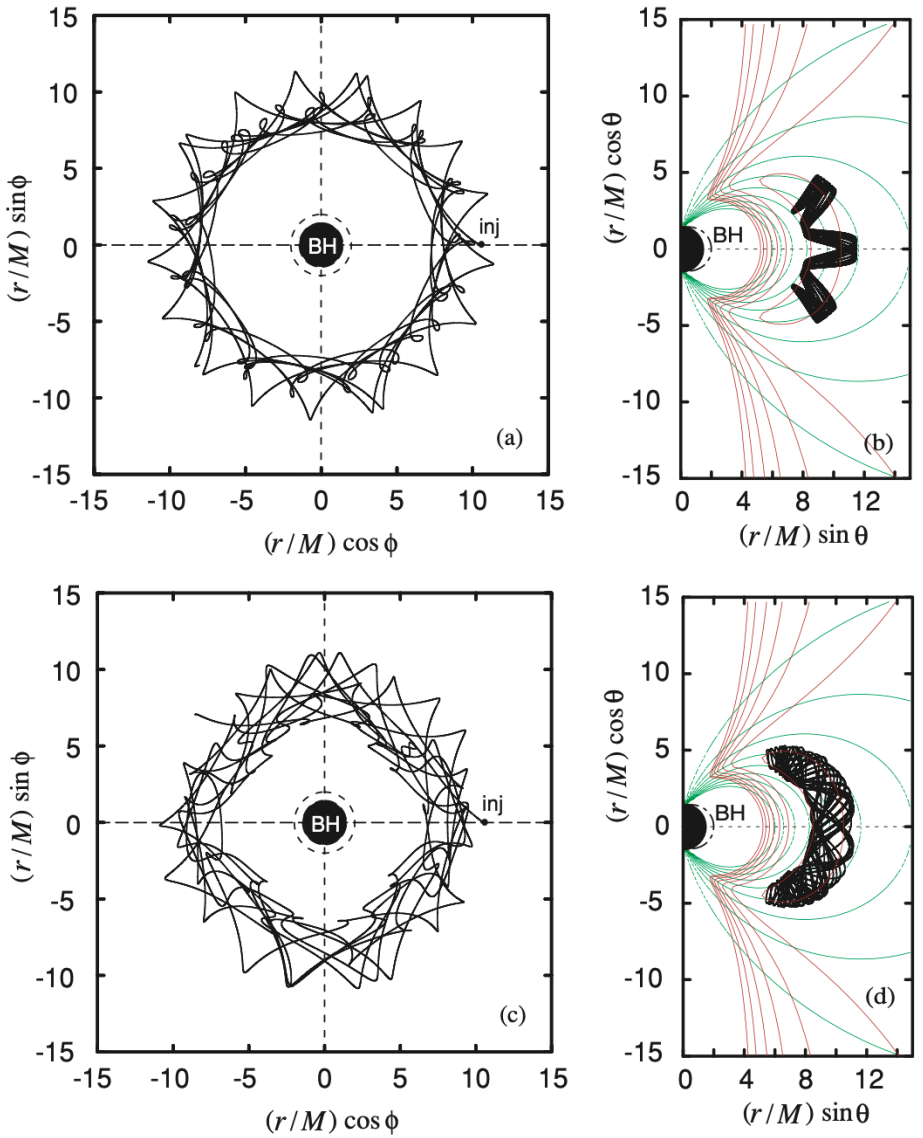


Figure: Reproduced from Fig. 6 of [9]

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